during the initial handshake, each of which needs different cryptographic primitives to be implemented. Some of these also require digital signatures to be implemented, again with several choices of algorithms. Some signature algorithms support different group parameters, such as elliptic curves used for ECDSA signatures, which require further code. After the handshake completes, there are several choices for cipher modes and MAC algorithms for securing application data. X.509 certificate authentication itself requires additional code. This can add up to a significant amount of code to include on a constrained device.